Real Estate Mobile Application

- **Prepared by:** Michael Tanios – Sedhom Samy
- **Course:** Wireless and Sensor Networks (IT304)
- **Instructor:** Eng. Ahmed Elkhatib
- **Department :** IT

- **Academic Year:** 2026

1. Executive Summary

This proposal presents a mobile real estate platform designed to help people in Egypt find homes for rent or purchase and help owners publish property listings. Users can filter available properties by transaction type, governorate, city or district, budget, and property characteristics. Owners can create listings with photographs, location details, pricing, and amenities.

The first release will focus on reliable listings, clear search, and convenient in-app inquiries and viewing requests. Its central differentiator is listing reliability: listings expire automatically unless owners reconfirm availability, and phone verification and moderation help limit duplicate and misleading listings. The service will be free for both property owners and users during the initial growth phase. Future revenue options may include promoted listings, professional account subscriptions, and relevant advertising, introduced only after the platform has built meaningful usage.

The proposed implementation uses Flutter for a cross-platform mobile application and Supabase for authentication, PostgreSQL data storage, image storage, and backend services. The platform is intended to support all Egyptian governorates, while its operational launch will begin in one pilot governorate or area and expand in stages as local listing coverage reaches agreed targets.

2. Problem Statement

People searching for a property often need to compare information spread across social media pages, informal groups, and multiple listing sources. Listings may be incomplete, outdated, difficult to filter, or unclear about location and price. Owners also need a simple way to present property details and receive organized inquiries.

The proposed application addresses these challenges by bringing property discovery and listing management into one searchable mobile experience. It gives users structured filters and gives owners a consistent format for presenting properties.

3. Project Aim and Objectives

Aim: To design and implement a user-friendly mobile application that connects property owners with people looking to rent or buy real estate in Egypt.

- Allow owners to create, edit, pause, and manage property listings.
- Allow users to search and filter listings by rent or sale, location, budget, and property attributes.
- Present property information in a consistent format with photographs, description, price, and location.
- Enable users to save favorite listings and submit inquiries or viewing requests.
- Provide an administrative workflow for reviewing reported or potentially misleading listings.
- Keep listings current through automatic expiry, owner reminders, and one-tap reconfirmation of availability.
- Build trust through phone-verified accounts, review of new listings before publication, and flags for possible duplicate listings. Phone verification confirms access to a reachable phone number; it does not establish legal ownership.
- Build the first version with a free access model and a technical foundation that can support future revenue features.

4. Intended Users and Roles

Role	Primary needs
Property seeker	Search listings, compare options, save favorites, and send an in-app inquiry or request a viewing.
Property owner	Publish accurate property details, upload photos, receive inquiries, and manage listing status, including confirming availability.
Administrator	Review reports, moderate listings, manage categories and locations, and respond to support issues.

5. Proposed Scope

5.1 Minimum Viable Product

- Account registration and sign-in.
- Property listing creation with transaction type, property type, price, description, location, amenities, and multiple photos.
- Search and filtering by rent or sale, governorate, city or district, price range, property type, bedrooms, and area.
- Search results and a detailed property page.
- Favorites and in-app inquiry or viewing-request flow.
- Owner dashboard for managing active, paused, and completed listings.
- Listing freshness: a configurable expiry period, reminders before expiry, one-tap reconfirmation of availability, and a visible “last confirmed” date. A 30-day period will be tested as the initial pilot setting and adjusted based on owner response and listing freshness.
- Phone number verification by one-time code (OTP) before an owner’s first listing is published. This confirms a reachable owner, not legal ownership.
- In-app inquiries and viewing requests are the only contact channels in the first release. Direct phone and WhatsApp contact are excluded from the MVP so inquiries and viewing coordination remain within the application.
- Basic notifications for new inquiries, viewing requests, and upcoming listing expiry.
- Basic reporting and administrator moderation tools, including review of listings from new owners before publication.
- Arabic-first user interface with full right-to-left (RTL) layout, and an English interface as a possible follow-up depending on project requirements.

5.2 Out of Scope for the First Release

- Online rent collection, escrow, or payment processing.
- Legally binding digital lease or sale contracts.
- Automated property valuation.
- Guaranteed verification of title ownership or legal status.
- Paid subscriptions, promoted listings, and commercial advertisements.

6. Core User Workflows

6.1 Property Owner

1. Create an account and complete a basic profile, including phone verification by one-time code.
2. Select “Rent” or “Sell” and enter property details.
3. Choose a governorate, then a city or district, and optionally pin the location on a map.
4. Upload photographs and review the listing preview.
5. Submit the listing for moderation; listings from new owners are reviewed before they are published.
6. Manage inquiries and update the listing as available, paused, or no longer available. Reconfirm availability when reminded; listings that are not reconfirmed expire automatically.

6.2 Property Seeker

7. Choose whether to rent or buy.
8. Select a governorate and optionally narrow the search to a city or district.
9. Set a budget and any relevant property filters.
10. Review results, check when each listing was last confirmed, and open property details.
11. Save a listing or send an inquiry/request a viewing through the application.

7. Functional Requirements

ID	Requirement
FR-01	The system shall allow a user to register, sign in, and manage a basic profile.
FR-02	The system shall allow an owner to create, edit, pause, and close a property listing.
FR-03	A listing shall include transaction type, property type, price, location, description, and property attributes.
FR-04	The system shall support multiple photographs per property listing.
FR-05	The system shall allow seekers to search by transaction type, location, and price range.
FR-06	The system shall provide optional filters such as bedrooms, bathrooms, and area.
FR-07	A seeker shall be able to save and remove favorite listings.
FR-08	A seeker shall be able to submit an inquiry or viewing request for a listing. A viewing request shall include a preferred date and time, and the owner shall be able to accept or decline it.
FR-09	Users shall be able to report a listing for administrator review.
FR-10	Administrators shall be able to review listings and reports and change listing status.

ID	Requirement
FR-11	Each listing shall have a configurable expiry period. A 30-day period will be tested during the pilot. The system shall remind the owner before expiry and automatically hide the listing if it is not renewed.
FR-12	An owner shall be able to confirm that a listing is still available with a single action, which updates its last-confirmed date and extends its expiry.
FR-13	Listings shall display their last-confirmed date, and seekers shall be able to sort results by most recently confirmed.
FR-14	The system shall verify an owner's phone number with a one-time code before the owner's first listing can be published. This verifies a reachable phone number, not legal ownership of the property.
FR-15	Listings from new owners shall be reviewed by an administrator before publication, and the system shall flag possible duplicates, such as repeated listings from the same phone number with the same location and price, for administrator review.
FR-16	Seekers and owners shall communicate through in-app inquiries and viewing requests. Direct phone and WhatsApp contact are excluded from the MVP.
FR-17	The system shall notify owners of new inquiries and viewing requests and of upcoming listing expiry.
FR-18	Administrators shall be able to add and edit governorates, cities or districts, and neighborhoods in the location catalogue.

8. Non-Functional Requirements

- Usability: Common tasks should be understandable on a mobile screen with clear labels and simple navigation.
- Performance: Search results and listing details should load promptly under normal network conditions. Database indexes on transaction type, governorate, city, and price should support the most frequent search queries.
- Security: Access to user and listing data should follow least-privilege rules; users may modify only records they own.
- Privacy: The application should collect only information needed to operate its features and explain how contact details are used. Owner phone numbers are used for account verification and are not displayed to seekers in the MVP. Inquiries and viewing coordination take place inside the application.
- Reliability: The application should handle failed uploads and network interruptions without losing entered listing information where practical.
- Media: Photos should be compressed on the device before upload, with limits on the number and size of photos per listing, to control storage cost and loading time.

- Scalability: Location data and search design should support expansion across all governorates and increasing listing volume.
- Maintainability: The mobile client, backend data model, and administrative functions should be organized so new features can be added safely.

9. Data Model Overview

The following entities provide a starting point for the relational database design. Exact fields and constraints will be refined during implementation.

Entity	Example information
Profiles	User ID, display name, phone or email, phone verification status, account role, created date.
Properties	Owner ID, transaction type, property type, price, description, bedrooms, bathrooms, area, listing status, expiry date, last-confirmed date.
Locations	A hierarchical catalogue (governorate → city or district → neighborhood), each level with Arabic and English names, parent ID, and status; optional street-level details, latitude, and longitude on each property.
Property photos	Property ID, storage path, display order, upload timestamp.
Favorites	User ID, property ID, saved timestamp.
Inquiries	Property ID, seeker ID, message, request type, preferred viewing date and time, status, timestamps.
Reports	Reporter ID, property ID, reason, review status, administrator notes.
Notifications	User ID, type (inquiry, viewing request, expiry reminder), related listing, read status, timestamp.

10. Technology and Architecture

Layer	Proposed technology	Purpose
Mobile application	Flutter with Dart	Build a shared Android and iOS application.
Authentication	Supabase Auth	Manage user registration and sign-in.
Database	Supabase PostgreSQL	Store profiles, listings, location references, favorites, inquiries, and reports.
Photo storage	Supabase Storage	Store and serve property photographs.
Backend security	Supabase Row Level Security	Restrict access so users can manage

Layer	Proposed technology	Purpose
		only permitted data.
Administration	Web-based admin dashboard on top of Supabase (a ready-made tool or a simple web interface)	Review new listings and reports, manage the location catalogue, and moderate content.
Scheduled jobs	Supabase scheduled jobs (database cron or Edge Functions)	Expire listings and generate owner reminders automatically.
Notifications	Push notification service such as Firebase Cloud Messaging, plus in-app notifications	Alert owners about inquiries, viewing requests, and expiring listings.
Maps (optional for MVP)	Google Maps or another map provider	Display approximate property location and support map selection, subject to API setup and cost review.

The mobile application will communicate with Supabase through authenticated APIs. Location values should use a structured hierarchy—governorate, city or district, and neighborhood—so users can filter reliably even when map services are unavailable.

11. Launch and Growth Approach

The product is intended for all governorates in Egypt. However, a staged operational launch is recommended: prepare the location catalogue for nationwide use, then begin in a single pilot governorate or area where owners can be recruited directly and listings checked closely, and expand only when agreed targets for active listings and activity are met. Expansion can follow as the number of active listings and user activity grows in each area.

During the initial launch, the application will be free for owners and seekers. The main goal is to build a useful supply of listings, learn which search filters matter most, and improve listing quality and response rates. Potential revenue features should be tested later and should provide clear optional value.

12. Future Revenue Options

Option	Possible value to the customer
Promoted listing	Increased visibility for a selected listing for a defined period.
Professional subscription	Tools for owners or agencies managing multiple listings, inquiries, and performance insights.
Relevant advertising	Clearly labeled promotions from services related to housing, such as moving, furnishing, or maintenance.
Optional service packages	Additional support such as photography or listing preparation, subject to operational feasibility.

These are future possibilities, not launch requirements. Pricing and timing should be based on usage data and customer feedback rather than assumed demand.

Promoted listings and professional subscriptions are the most realistic first experiments, because they depend on owner and agency demand rather than very large user numbers. Commercial advertising generally needs a large audience and is best tested last.

13. Risks and Mitigation

Risk	Mitigation approach
Insufficient listings in a location	Launch area by area, recruit owners early, and show listing freshness and availability.
Outdated or misleading listings	Expire listings automatically unless owners reconfirm availability, send reminders, show the last-confirmed date, and provide reporting and moderation tools.
Low user trust	Use clear listing fields, transparent contact flows, phone number verification (OTP), pre-publication review of new owners' listings, duplicate-listing flags, and responsive support.
Users complete transactions outside the app	Make in-app inquiries, viewing coordination, saved searches, and listing management genuinely useful; do not rely on technical restrictions alone. Keep inquiries and viewing coordination inside the application in the MVP, while making the in-app flow simple enough to be useful to both parties.
Location data inconsistency	Maintain a controlled governorate/city catalogue instead of relying only on free-text location entry.
Privacy or access-control issues	Minimize collected personal data, configure database access policies carefully, and review security before release.
Operational or legal obligations	Review applicable Egyptian business, data, and real-estate rules with qualified professionals before offering paid brokerage or transaction services.

14. Development Roadmap

Phase	Deliverables
0. Validation	Interviews with 10–20 owners and agents and with seekers in the pilot area, a clickable prototype tested with real users, and target values for the success measures.
1. Planning and design	User journeys, screen wireframes, location hierarchy, and database schema.
2. Core foundation	Flutter project, Supabase setup, authentication, and access policies.
3. Listings	Create and manage listings, image uploads, listing details, expiry and availability reconfirmation, and phone verification.

Phase	Deliverables
4. Discovery	Search, filters, results, favorites, inquiry/viewing requests, and basic notifications.
5. Administration and refinement	Moderation, reporting, usability improvements, release preparation, and a pilot launch in the first target area.
6. Growth features	Saved-search alerts, duplicate-photo detection, map enhancements, analytics, and optional monetization experiments.

15. Success Measures

The first release should be evaluated using product and marketplace measures rather than revenue, since access will initially be free.

- Number of active and recently confirmed listings.
- Share of active listings that owners reconfirm before they expire.
- Searches that produce at least one relevant result.
- Inquiry and viewing-request volume per active listing.
- Owner response rate and time to first response.
- Share of listings with complete details and usable photographs.
- Repeat usage by seekers and owners.
- Reports, listing removals, and user support issues.

Target values for these measures should be set before the pilot begins (for example, a minimum number of active listings and a minimum monthly reconfirmation rate) and reviewed at the end of the pilot.

16. Competitive Landscape and Proposed Differentiation

The core property-search concept already exists in Egypt. Established platforms provide listing search, location and price filters, photographs, and ways to contact sellers or agents. The project should therefore compete through a focused service experience and stronger local execution, rather than relying on the existence of a mobile app as its main advantage.

16.1 Existing Platforms

Platform	Publicly described capabilities
Aqarmap	Sale and rental listings, property details, photographs, map locations, and direct seller contact. [1]
Property Finder Egypt	Search and filters, saved listings and searches, notifications, and contact options such as phone, SMS, WhatsApp, or email. [2]
Bayut Egypt	Buy and rent search, property filters, and map-based discovery. [3]
Official Egyptian Real Estate Platform	Property discovery, inquiries, viewing requests, and connections between owners and verified agents. [4]

16.2 How This Project Plans to Stand Out

The project will aim to be better for a specific group of users: people searching in Egyptian governorates and local areas where they need current, clearly described options, and individual owners who want a simple way to publish and manage a listing.

The primary differentiator is listing reliability: current, owner-confirmed listings from accounts with verified phone numbers. Phone verification confirms a reachable owner, not legal ownership. The other points below support this goal.

- Local listing coverage: prepare a structured governorate, city or district, and neighborhood catalogue, then build meaningful listing supply in each launch area instead of treating nationwide coverage as a filter alone.
- Listing freshness: listings expire automatically unless owners reconfirm availability with one tap, and each listing shows when it was last confirmed, helping seekers distinguish current listings from stale ones.
- Listing trust: phone-verified accounts, review of new owners' listings before publication, duplicate flags, and an easy way to report problems. Phone verification confirms a reachable owner, not legal ownership.
- Simpler experience for individual owners: make listing creation, photo upload, inquiry handling, and status updates straightforward without requiring a professional agency workflow.
- Organized viewing requests: let seekers request a viewing and let owners manage those requests in the app, creating a useful follow-up step after search.
- Free launch: keep core listing and search features free while the platform builds inventory, user trust, and activity; consider optional paid promotion or professional tools only after validating demand. Because seekers can generally already search for free on existing platforms, this is a growth tactic rather than a lasting advantage.

These are proposed differentiators, not proven competitive advantages. The project should validate them through interviews with property seekers and owners, a small pilot with real listings, and measures such as listing freshness, owner response rate, relevant search results, and completed viewing requests.

Sources

[1] Aqarmap, Google Play application description: https://play.google.com/store/apps/details?id=com.aqarmap.android

[2] Property Finder Egypt, mobile application features: https://www.propertyfinder.eg/blog/en/mobile-apps/

[3] Bayut Egypt: https://www.bayut.eg/

[4] Official Egyptian Real Estate Platform, terms and service scope: https://realestate.gov.eg/en/terms-and-conditions

17. Conclusion

The proposed project is a cross-platform real estate application for Egypt that organizes property discovery and owner listings in one mobile service. Flutter and Supabase provide a practical foundation for the first version, while structured location data and clear listing workflows support national expansion. The project's central bet is that current, verified listings, tested first in one pilot area, will earn more trust than larger but staler inventories; this must be validated with real users. A free launch allows the project to focus on inventory, trust, and user activity before testing optional revenue features.